UNO R4 WiFi OTA Setup Guide

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Overview

This guide sets up Arduino IDE 2 on a Windows laptop to upload sketches to an UNO R4 WiFi over WiFi, with no USB cable after the first upload. It takes about 15 minutes.

The board package has no built-in WiFi upload, so three pieces are added: the ArduinoOTA library on the board, the `arduinoOTA` upload tool on the PC, and a `platform.local.txt` file that tells the IDE how to use that tool.

Versions this was tested with:

Component	Version
Arduino UNO R4 Boards (`renesas_uno`)	1.6.0
ESP32 bridge firmware	0.6.0
ArduinoOTA library (Juraj Andrássy)	1.1.1
arduinoOTA upload tool (from Arduino AVR Boards)	1.3.0

Step 1: Update the board package and bridge firmware

The board package and the ESP32 bridge firmware must match, or WiFi features misbehave.

1. In the IDE, open **Tools → Board → Boards Manager**, search **UNO R4** and install the latest **Arduino UNO R4 Boards**.
2. Plug in the board over USB and run **Tools → Firmware Updater**. Select the UNO R4 WiFi and install the latest firmware.
3. Unplug and replug the board when it finishes.
4. Check the versions with this sketch (Serial Monitor at 115200). Both lines must show the same version.

```
#include <WiFiS3.h>

void setup() {
  Serial.begin(115200);
  while (!Serial) {}
  Serial.print("Installed bridge firmware: ");
  Serial.println(WiFi.firmwareVersion());
  Serial.print("Version this board package expects: ");
  Serial.println(WIFI_FIRMWARE_LATEST_VERSION);
}

void loop() {}
```

If the Firmware Updater can't see the board, reflash the bridge by hand: download the latest release of `arduino/uno-r4-wifi-usb-bridge` from GitHub, connect the **DOWNLOAD** pin to **GND** on the 2×3 ESP header next to the USB-C port (labels are on the back of the board), plug in USB, run `update.bat`, then remove the jumper and replug.

Step 2: Install the library and upload tool

1. **Tools → Manage Libraries**, search **ArduinoOTA** and install the one **by Juraj Andrássy**. Not "ArduinoOTA by Arduino", which is for other boards.
2. **Tools → Board → Boards Manager**, install **Arduino AVR Boards**. You won't use AVR boards; this package is what installs the `arduinoOTA` upload tool.
3. Check the tool is there. This folder should exist:

```
%LOCALAPPDATA%\Arduino15\packages\arduino\tools\arduinoOTA
```

Step 3: Add platform.local.txt to the board package

This file adds WiFi upload to the IDE. Create it in the board package folder, next to `platform.txt`. Replace `1.6.0` with your installed version:

```
%LOCALAPPDATA%\Arduino15\packages\arduino\hardware\renesas_uno\1.6.0\platform
.local.txt
```

Paste the address above (without the file name) into File Explorer's address bar, since `AppData` is hidden. Create the file with this content:

```
# This configuration file supports the general ArduinoOTA library
https://github.com/jandrassy/ArduinoOTA

#IDE 1
tools.bossac.network_cmd={runtime.tools.arduinoOTA.path}/bin/arduinoOTA
tools.bossac.upload.network_pattern="{network_cmd}" -address {serial.port} -
port 65280 -username arduino -password "{network.password}" -sketch "
{build.path}/{build.project_name}.bin" -upload /sketch -b

tools.dfu-util.network_cmd={runtime.tools.arduinoOTA.path}/bin/arduinoOTA
tools.dfu-util.upload.network_pattern="{network_cmd}" -address {serial.port}
-port 65280 -username arduino -password "{network.password}" -sketch "
{build.path}/{build.project_name}.bin" -upload /sketch -b

#IDE 2
upload.tool.network=arduino_ota
tools.arduino_ota.cmd={runtime.tools.arduinoOTA.path}/bin/arduinoOTA
tools.arduino_ota.upload.pattern="{cmd}" -address "{upload.port.address}" -
port 65280 -username arduino -password "password" -sketch "
{build.path}/{build.project_name}.bin" -upload /sketch -b

## arduinoOTA as programmer. add entries with {ip} into programmers.txt
tools.arduinoOTA.cmd={runtime.tools.arduinoOTA.path}/bin/arduinoOTA
tools.arduinoOTA.program.params.verbose=
tools.arduinoOTA.program.params.quiet=
tools.arduinoOTA.program.pattern="{cmd}" -v -address {ip} -port 65280 -
username arduino -password password -sketch "
{build.path}/{build.project_name}.bin" -upload /sketch -b
```

This is the library's `extras\renesas\platform.local.txt` with one change: the IDE 2 upload line sends the password `password` directly instead of asking for it. IDE 2 remembers the first password typed and never asks again, so a typo there causes `Unauthorized` on every upload. If you change the password in your sketch, change it in this line too.

Then **close and reopen the IDE**. Repeat this step after every update of the UNO R4 board package, because the update creates a new version folder without this file.

Step 4: Upload the OTA sketch over USB (once)

Create a sketch named `UnoR4OTA` with two tabs. Upload it over **USB** the first time; after that, every upload can go over WiFi.

`arduino_secrets.h` (new tab, your own WiFi details):

```
#define SECRET_SSID "YourWiFiName"
#define SECRET_PASS "YourWiFiPassword"
```

`UnoR4OTA.ino`:

```
#include <WiFiS3.h>
#include <ArduinoOTA.h>
#include "arduino_secrets.h"

// Change this between uploads to see that an OTA update worked
const unsigned long BLINK_INTERVAL = 1000;   // ms

unsigned long lastBlink = 0;
bool ledOn = false;

void setup()
{
    Serial.begin(115200);
    delay(2000);

    pinMode(LED_BUILTIN, OUTPUT);

    Serial.println("[APP] Connecting to WiFi...");
    while (WiFi.begin(SECRET_SSID, SECRET_PASS) != WL_CONNECTED)
    {
        delay(2000);
    }

    // WiFiS3 can report connected before DHCP finishes, so wait for a real
IP
    Serial.println("[APP] Waiting for IP address...");
    unsigned long start = millis();
    while (WiFi.localIP() == IPAddress(0, 0, 0, 0))
    {
        if (millis() - start > 15000)
        {
```

```
            Serial.println("[APP] No IP after 15 s, retrying WiFi...");
            WiFi.disconnect();
            delay(1000);
            WiFi.begin(SECRET_SSID, SECRET_PASS);
            start = millis();
        }
        delay(250);
    }

    Serial.print("[APP] Connected, IP: ");
    Serial.println(WiFi.localIP());

    // name shown in the IDE, upload password
    ArduinoOTA.begin(WiFi.localIP(), "UnoR4WiFi", "password",
InternalStorage);
    Serial.println("[APP] OTA ready");

    Serial.print("[APP] Blink interval: ");
    Serial.print(BLINK_INTERVAL);
    Serial.println(" ms");
}

void loop()
{
    ArduinoOTA.poll();   // must run often, so no delay() here

    // Non-blocking blink
    unsigned long now = millis();
    if (now - lastBlink >= BLINK_INTERVAL)
    {
        lastBlink = now;
        ledOn = !ledOn;
        digitalWrite(LED_BUILTIN, ledOn ? HIGH : LOW);
    }
}
```

After the upload, the Serial Monitor (115200) should show a real IP such as `192.168.254.106`, then `[APP] OTA ready`.

Rules for every sketch you upload over WiFi from now on:

- Keep `ArduinoOTA.begin(...)` in `setup()` and `ArduinoOTA.poll()` in `loop()`. A sketch without them can only be updated over USB again.

- Keep `loop()` free of long `delay()` calls; use `millis()` timing as the blink does, or uploads time out.
- Keep the wait-for-IP loop; without it the board can start OTA on `0.0.0.0` and never appear on the network.

Step 5: Upload over WiFi

1. With the OTA sketch running, open **Tools → Port → Network ports** and pick **UnoR4WiFi at 192.168.x.x**. It can take 30–60 seconds to appear.
2. Change `BLINK_INTERVAL` to `200` and click **Upload**. No password box appears; the password comes from `platform.local.txt`.
3. The output should end with `Flashing sketch ... done` and no error. The **L** LED next to pin 13 now blinks fast.

If the network port never appears (some routers or Windows Firewall block mDNS), upload by IP from PowerShell. Click **Verify** first so the `.bin` is current, then use the board's IP and your sketch's build folder (shown in the verbose compile output):

```
& (Get-ChildItem
"$env:LOCALAPPDATA\Arduino15\packages\arduino\tools\arduinoOTA" -Recurse -
Filter arduinoOTA.exe | Select-Object -First 1).FullName -address
192.168.254.106 -port 65280 -username arduino -password password -sketch "
<build folder>\UnoR4OTA.ino.bin" -upload /sketch -b
```

The PC and board must be on the same network. Set the Windows network profile to **Private** and allow the Arduino IDE through the firewall when asked.

Troubleshooting

Symptom	Cause	Fix
USB upload hangs after a firmware update	Board left in ESP download mode, COM port changed, or board package older than the bridge firmware	Unplug and replug, reselect the port, update Arduino UNO R4 Boards

Symptom	Cause	Fix
`OTAUpdate` sketch stops at "Applying update..." with `uart driver error`	Normal: the ESP32 is reflashing the main chip, so the old sketch can't print anymore	Check the board's behaviour (the LED matrix animates for the Arduino demo file)
Serial shows `IP: 0.0.0.0`	`WiFi.begin()` returned before DHCP gave an address	Keep the wait-for-IP loop from Step 4
`undefined reference to 'loop'`	The sketch has no `loop()` function	Add `loop()` with `ArduinoOTA.poll()`
`Failed uploading: no upload port provided`	Missing `platform.local.txt` or `arduinoOTA` tool, so no network port	Do Steps 2 and 3, restart the IDE
`Error flashing the sketch:Unauthorized`	Upload password doesn't match the one in `ArduinoOTA.begin()` (it's case-sensitive)	Make the password in `platform.local.txt` match the sketch, restart the IDE
Board seems dead after a failed update	A partial sketch was written	Double-tap RESET, select the port, upload over USB
Hard to read logs among ESP32 output	The bridge firmware prints its own lines to the same port	Prefix your logs with `[APP]` and filter: `arduino-cli monitor -p COM5 -c baudrate=115200 | Select-String "\[APP\]"`